# EXPERIMENT 14.1.1

## ALGORITHM: Addition of Two Complex Numbers using Class

Step 1: Start

Step 2: Define a class named Complex

Step 3: Define __init__() method

  Initialize:

    real = 0

    imag = 0

Step 4: Define initComplex() method

  Read input from user

  Split input into two values

  Convert values to integers

  Assign:

    real = first value

    imag = second value

Step 5: Define sum(c1, c2) method

  self.real = c1.real + c2.real

  self.imag = c1.imag + c2.imag

Step 6: Define display() method

  If self.imag >= 0

    Print: real + imag i

  Else

    Print: real - |imag| i

Step 7: In main program

  Create object c1

  Create object c2

  Create object result

Step 8: Call c1.initComplex()

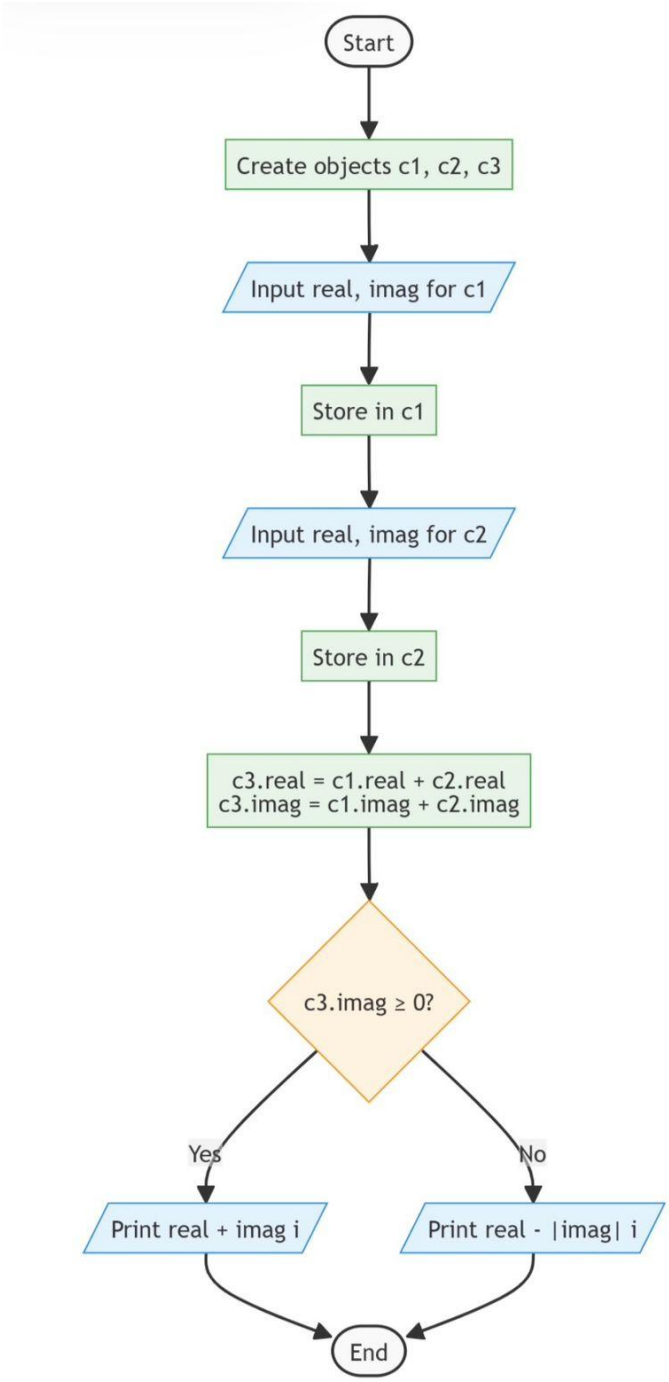
Step 9: Call c2.initComplex()

Step 10: Call result.sum(c1, c2)

Step 11: Call result.display()

Step 12: Stop

**FLOWCHART:**



## CODETANTRA:

CODETANTRA Home ira.handa.batch2025@sitnagpur.siu.edu.in Support Logout

**14.1.1. Complex Number Addition using Class** 09:17

Write a Python program using a class named **Complex** to add two complex numbers.

Create a class **Complex** with the following methods:

- **initComplex()**: Read user input for the real and imaginary parts.
- **sum()**: Compute the sum of two complex numbers and store the result in the current instance.
- **display()**: Display the complex number in the format "<a> + <b>i" or "<a> - <b>i".

The program should create three instances of **Complex** class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.

**Input Format:**

- First line contains two space-separated numbers: real and imaginary parts of first complex number
- Second line contains two space-separated numbers: real and imaginary parts of second complex number

Explorer

complex... Submit

```python
class Complex:
    def __init__(self):
        self.real = 0
        self.imag = 0
    def initComplex(self):
        line = input().split()
        if line:
            self.real = int(line[0])
            self.imag = int(line[1])
    def sum(self, c1, c2):
        self.real = c1.real + c2.real
        self.imag = c1.imag + c2.imag
    def display(self):
        if self.imag >= 0:
            print(f"{self.real} + {self.imag}i")
        else:
            print(f"{self.real} - {abs(self.imag)}i")
```

Terminal Test cases

Debugger